

Supplementary Information for

When do single-cell response distributions improve drug retrieval?

This Supplementary Information describes the methods, additional analyses and results supporting the main text. It contains Supplementary Methods, four Supplementary Notes and four Supplementary Tables. The supplementary analyses follow the same conceptual structure as the main text: they examine the information retained by population-level retrieval, the dependence of retrieval gains on the evaluation criterion, and the conditions under which state-specific single-cell responses can become consequential for candidate ranking.

10 Supplementary Methods

11 **Implementation of population retrieval scores.** Energy distance was the primary population score.
12 RBF-kernel maximum mean discrepancy used the median squared pairwise distance of the pooled
13 query and candidate sample as its bandwidth reference and evaluated scales of 0.25, 1 and 4 times
14 that value. Sliced Wasserstein-1 distance averaged quantile-matched projections over 128 fixed ran-
15 dom directions. Candidate populations were subsampled with a fixed seed to at most 500 cells in the
16 observed-population analyses and 300 cells in the partial-observation analyses. A population containing
17 fewer than two cells after filtering was assigned a fixed large penalty distance rather than removed from
18 the candidate library. Mean cosine and mean L_2 first reduce each treated population to its mean and
19 subtract the corresponding matched control mean; query and candidate controls may differ. Energy dis-
20 tance, RBF-MMD, sliced Wasserstein and coverage receive raw normalized-expression cell populations
21 and do not subtract matched controls in the manuscript population-scoring path. Energy and RBF-
22 MMD use the unbiased U-statistic by default; sliced Wasserstein has no U/V choice, and coverage uses
23 the same U default for its energy/MMD component distances. The PCA-latent population diagnostic is
24 the explicitly labelled legacy V-statistic exception.

25 **Implementation of subpopulation coverage.** In construction-labelled experiments, the known construc-
26 tion labels are used directly when they are carried by the task. In the controlled-mixture ranker, candi-
27 date labels are unavailable, so candidate cells are assigned to the nearest target-component mean by
28 Euclidean distance. For partial observation, query cells are partitioned by k -means with $k = 2$, three
29 initializations and a fixed seed; candidate cells are assigned to the query component whose centroid
30 has the highest cosine similarity after row normalization. A candidate with no cell assigned to a query
31 component received the penalty distance for that component. At small β , the coverage distance was eval-
32 uated using $D_\beta = \text{mean}(d) + \frac{\beta}{2} \text{var}(d) + O(\beta^2)$ to avoid numerical cancellation in the log-sum-exp form.
33 Coverage-mean and coverage-worst use the $\beta \rightarrow 0$ and $\beta \rightarrow \infty$ limits, respectively. The base distance
34 was energy distance unless otherwise specified; the complete score registry is given in Supplementary
35 Table 1.

36 Supplementary Note 1. Additional diagnostics of when population information changes 37 ranking

38 The main text separates the existence of state-specific perturbation responses, their recoverability and
39 their effect on candidate ordering. Additional analyses evaluated whether simple observables could
40 identify queries for which population-level retrieval is more useful.

41 **Control-reference sensitivity.** The primary population scores use raw normalized-expression popula-
42 tions, whereas the mean scores use control-referenced means. We repeated the existing cross-line rank-
43 ings after subtracting the matched control mean from each population, retaining the query, candidate
44 library, seeds, distance functions and evaluation unchanged. Across 2,250 SciPlex3 cross-line queries,
45 energy retrieval changed from Hit@1 0.9189 to 0.9158 and MRR 0.9478 to 0.9462; sliced Wasserstein
46 changed from Hit@1 0.8123 to 0.8075 and MRR 0.8768 to 0.8741. RBF-MMD and both coverage limits
47 were unchanged to the reported precision, and every population-minus-mean effect remained positive
48 in each cross-line setting. The Frangieh IFNGR1 contrasts were unchanged. Controlled-mixture, GDSC2
49 and same-condition RNA rankings use a common control translation, which leaves these population dis-
50 tances unchanged. For partial observation, rerunning query clustering and candidate assignment after
51 control subtraction changed the rankings: across all 765 queries, the mean regret reduction for coverage-
52 worst changed from +0.1320 to +0.2021 and for coverage-mean from +0.1155 to +0.2041, while the
53 corresponding mean minority-coverage gains changed from +0.0027 to +0.0057 and from +0.0021 to
54 +0.0059; the assignment changed for essentially all candidates and the top-ranked candidate changed in
55 88.5% and 86.7% of queries, respectively. The response-matching advantage therefore remained positive,
56 but the rankings were not invariant.

Query identifiability. In A549 leave-drug-out retrieval, query identifiability was quantified as a two-cluster silhouette on the query cells, independently of the retrieval outcome. Across 543 evaluable query-seed combinations it was weakly and non-monotonically associated with the absolute mechanism-of-action nDCG of a single energy-distance run (Spearman $\rho = 0.17$, $p = 6.3 \times 10^{-5}$), and the association is confounded with candidate-library composition: the highest-identifiability tertile holds a median of six same-mechanism candidates against two in each lower tertile, and a larger relevant pool mechanically raises the attainable nDCG. The median silhouette score even in that tertile is 0.038, so the two-state structure these queries carry is weakly resolved. This analysis scores one retriever and therefore bears on absolute recovery rather than on the population-minus-mean gain.

Unsupervised recovery of constructed two-state structure. Response-identity recoverability in the constructed two-state mixture is a supervised ceiling against the best unsupervised method (ceiling 0.692 against best unsupervised 0.674), and separately a ceiling of 0.879 when the same cells are posed at the level of individual drug pairs rather than pooled drug classes. Three further measurements characterise the same limit.

First, across nine unsupervised clustering configurations, the median adjusted Rand index against the known two-source labels ranged from 0.037 to 0.106. The configurations were k -means with $k = 2$ in the raw, 10-dimensional PCA, 50-dimensional PCA and response spaces, two-component Gaussian mixtures in the raw and 50-dimensional PCA spaces, and k -means with k selected from 3, 4 and 5 by silhouette, reported separately by which k was selected. The highest value, 0.106, was obtained when silhouette selection chose $k = 4$. Recovery is weak under all nine, so the low ceiling is not specific to the clustering method used in the main analysis.

Second, the weak separation is visible in the raw geometry as well as in an accuracy score. Silhouette scores computed independently for each perturbation-response population under a two-cluster partition had a median of 0.034 in SciPlex3 ($n = 529$ populations) and 0.084 in Frangieh ($n = 231$); in both datasets fewer than 0.5% of populations exceeded 0.1. Visible multimodality was not required for recoverable state-specific perturbation information.

Third, recovery is determined by state separation. Averaging the k -means $k = 2$ adjusted Rand index over 20 seeds across a grid of source separation and cells per source, increasing the cell budget from 25 to 400 cells per source changed recovery by at most 0.02 at fixed separation, whereas increasing separation from 1.0 to 8.0 times the observed value raised it from 0.07 to 1.00. The separation transform was defined from the full source populations before subsampling. Increasing cell number did not compensate for closely separated states, consistent with the supervised ceiling.

Measured response divergence. We stratified the partial-observation queries by measured response divergence, one minus the mean cosine between their state-specific perturbation responses. For minority-state coverage, the formal individual-drug quartile means were -0.0035 , $+0.0030$, $+0.0038$ and $+0.0036$ (Q1–Q4), with drug-cluster bootstrap 95% intervals of $[-0.0091, +0.0013]$, $[+0.0017, +0.0045]$, $[+0.0025, +0.0052]$ and $[+0.0021, +0.0052]$; the cluster-bootstrap Spearman association was $\rho = +0.1152$ (95% CI $[+0.0147, +0.2143]$). Under mechanism recovery, the corresponding means were -0.0497 , -0.0391 , -0.0158 and $+0.0142$; the Q4 interval remained $[-0.0165, +0.0425]$, so the highest-divergence stratum did not support a positive mechanism-recovery advantage. The mechanism-divergence association was $\rho = +0.1548$ (95% CI $[+0.0722, +0.2349]$). The two least-divergent mechanism strata had BH-adjusted drug-level $q = 0.140$; Q3 and Q4 intervals crossed zero.

These results distinguish differential response from decision relevance. Stronger state-specific responses provide more information for population-aware retrieval, but candidate ordering can remain unchanged.

Precision of the two task-proximal comparisons in the highest-divergence stratum. In the most divergent quartile the two criteria differ in precision rather than effect size. The mechanism-recovery gap is nearly three times the minority-coverage gap but its per-query standard deviation is twelve times larger (0.169 against 0.0136), so the two readings are asymmetric: the minority-state comparison returns a small

effect, whereas the mechanism-recovery comparison remains uncertain in this stratum. Achieved power and the sample size required for 80% power are given in the code release. The main text’s mechanism-recovery conclusion uses the full 600-query set and the shape of the whole distribution (Fig. 3a,b).

Instability of thresholded dominance counts. Thresholded dominance counts were unstable under resampling: at a gain threshold of +0.01, only 2 of the 10 initially threshold-crossing cross-line tasks stay above it in at least 8 of 10 seed resamples, while control drugs that cross it in no recorded seed cross it in 1 to 4 of 10 resamples. The main text therefore reports continuous retrieval differences and stratified analyses.

Diagnostic group comparison. The diagnostic assigns 127 queries to “mean or no call” and a further 11 to “mean sufficient”. Pooling the 11 into the non-recommended group leaves the verdict unchanged, and the formal individual-drug comparison of recommended versus no-call queries gives a recommended-minus-no-call median regret-gain difference of -0.0109 (95% drug-cluster bootstrap CI $[-0.0355, +0.0498]$; 584 versus 125 queries from 141 drugs). This comparison does not enrich for population-beneficial queries. The 11 mean-sufficient queries remain a descriptive verdict group.

Supplementary Note 2. HIR-Bench construction and biological transfer

HIR-Bench is a controlled construction in which the conditions under which within-population structure can alter candidate preference are known analytically. It tests the logic of a preference-changing regime; the analytical boundary is not a biological phase boundary.

The generator satisfies its internal consistency checks on the full parameter grid. When the two subpopulations have no conflict in candidate preference, the retrieval-flip rate is 0; the median distance between the grid’s own mixing fraction and the analytical boundary

$$\alpha^* = \frac{B}{A + B},$$

is 0.292 within the unit interval; and under a predicted-mean information condition the population-level and mean-level Hit@1 differ by 0.051 against a threshold of 0.10, which is the degeneracy expected when the retriever receives no within-population structure.

The measured counterpart of the analytical boundary behaves as the construction predicts. Over the 36 cells of the conflict-by-majority-fraction grid, the energy distance exceeds mean-signature retrieval in Hit@1 in 30 of them, and the advantage runs from -0.25 to $+1.00$, largest where conflict is high and the majority fraction is above α^* and vanishing below it.

We evaluated whether the synthetic regime could be recognized from quantities observable at retrieval time. Observable features included the subpopulation-variance ratio, isotropy index and response diversity of the query and candidate populations. With the 28 label-determining parameter cells as the cross-validation units, the pooled out-of-fold AUC was 0.788 against a majority-class rate of 0.643. Splitting on the 672 grid configurations gave 0.836; the resulting increase reflects near-duplicate configurations across folds. Because the label is deterministic within a parameter cell, the pooled out-of-fold AUC is the analysis statistic. Removing within-population information also increased decision regret, which a predicted-structure condition recovered.

The analytical boundary was also projected to the biological retrieval tasks as a transfer test. This projection did not generalize. The resulting labels were strongly imbalanced (229 versus 10), for which a constant majority-class predictor achieves 95.8% accuracy. The synthetic-regime projection reached 39.7%, far below this baseline and below it on four of the five datasets separately (0.29, 0.40, 0.33 and 0.22); the fifth is the predicted-mean subset, where the projection returns the same verdict for every task by construction. The boundary was transferred by percentile rather than by value: the crossover percentile was fitted on 4,480 grid cells of the HIR-Bench method-performance layer and applied to the real tasks at the corresponding quantile of their own preference-conflict distribution.

150 HIR-Bench establishes constructively that within-population structure can become consequential for
151 candidate choice. The corresponding regime in biological data requires direct biological measurement.

152 **Supplementary Note 3. Robustness of the natural-heterogeneity and large-scale analyses**

153 The main text uses patient-derived glioblastoma and Tahoe-100M to ask whether the information re-
154 quirements identified in the controlled experiments are also present when cellular heterogeneity arises
155 within biological samples. These datasets measure differential response, recoverability and the extent
156 to which state-specific response ordering leaves headroom for population structure to alter candidate
157 preference.

158 **Response alignment between compartments.** For 17 patient–drug pairs in which both compartments
159 carried enough treated and control cells, the cosine between the malignant and myeloid perturbation-
160 response directions has a median of 0.566, against a range of 0.02 to 0.39 for the constructed mixtures
161 used elsewhere in this study. Read directly, that contrast says the two compartments of a real tumour
162 respond far more alike than two experimentally combined cell lines do.

163 The cosine is descriptive and depends on effect size, signal-to-noise and cell number. The two
164 natural-tissue results in the main text, the recoverability gap and conservation of candidate ordering
165 across disjoint compartments, are measured independently of this cosine.

166 **Robustness of glioblastoma compartment assignment.** The ZhaoSims2021 dataset does not provide
167 resolved cell-type labels for the analysed cells, so malignant, tumour-associated myeloid and oligoden-
168 drocyte compartments were assigned from marker scores. Cells with fewer than 200 detected genes
169 were removed, counts were library-size normalized to 10^4 and $\log(1 + x)$ transformed, and gene sym-
170 bols were upper-cased; compartment scoring runs on that full matrix, before the 2,000-gene selection
171 used downstream, so the labels do not depend on a gene set chosen for downstream variance. Three
172 marker sets were used: malignant *SOX2*, *EGFR*, *PDGFRA*, *OLIG1*, *OLIG2*, *GFAP* and *AQP4*; myeloid
173 *PTPRC*, *CD14*, *AIF1*, *CSF1R*, *ITGAM*, *C1QA* and *C1QB*; oligodendrocyte *MBP*, *PLP1*, *MOG* and *MAG*. A
174 marker absent from the matrix is skipped. Each gene was z-scored across all cells of the cohort and the
175 compartment score is the mean of its set’s z-scores, so the scores are cohort-dependent by construction.
176 A cell was called for the highest-scoring compartment only if the mean log-normalized expression of
177 that compartment’s key markers (malignant *SOX2*, *OLIG2*, *EGFR*; myeloid *CD14*, *CSF1R*, *C1QB*, *AIF1*;
178 oligodendrocyte *MBP*, *PLP1*) reached 0.25 and its z-score margin over the second-highest compartment
179 reached 0.25. Cells failing either criterion were left unassigned, leaving 39.9% of cells unassigned. Ev-
180 ery called compartment was required to express its own key markers most strongly; this validation is
181 recorded in `results/zhao_gbm/compartment_validation.csv`.

182 The results were stable when both thresholds were varied jointly over

$$\{0.10, 0.25, 0.40, 0.50\}.$$

183 The unsupervised arm is the best of four partitions of the same cells: k -means with $k = 2$, a two-
184 component Gaussian mixture, Leiden¹ and HDBSCAN². Leiden is the maximum on 152 of the 180
185 natural splits, so the reported best-unsupervised value is close to Leiden’s alone.

186 Across this range, the excluded fraction varied from 29% to 58%, while the supervised drug-response
187 ceiling remained between 0.915 and 0.923, the best unsupervised recovery between 0.77 and 0.80, and the
188 paired recoverability gap between approximately 0.11 and 0.13. The median cosine between malignant-
189 and myeloid-specific perturbation-response directions remained between 0.54 and 0.59 across thresh-
190 olds.

191 **Patient-level uncertainty.** The natural-tissue recoverability analysis contains 36 within-patient drug-
192 pair splits across four patients and two compartments, with 30 of the 36 splits contributed by a single
193 patient. Drug-pair splits were therefore not treated as independent replicates. Patient-level cluster boot-
194 strap resampling was used to quantify uncertainty, and the analysis was repeated after removing the
195 dominant patient.

The reported recoverability gap is computed within each split before aggregation,

$$G_j = A_{\text{sup}}^{(j)} - A_{\text{unsup}}^{(j)},$$

not as the difference between the two marginal medians. The primary paired median is +0.117; subtracting the marginal medians would instead give +0.147. Across 5,000 patient-cluster bootstrap resamples, the natural-minus-constructed difference in recoverability gap was +0.109 with an interval of [+0.052, +0.129]. Four patient clusters contribute to this calculation, so the interval describes uncertainty within this cohort.

Two aggregation controls give the same gap. Fixing every probe-cluster pair, so that the same pair is scored in both arms, gives +0.122 in tissue against -0.002 in the constructed arm. Removing the patient that contributes 30 of the 36 splits gives +0.123, with a [+0.098, +0.138] interval.

Large-scale measurement in Tahoe-100M. Tahoe-100M plate 3 provides a substantially larger setting in which heterogeneity occurs within cell lines rather than through constructed mixtures. Two cellular-state definitions are used below, and both are fixed before any treated cell is examined. The first is the released cell-cycle phase call, G1 against G2M, with S-phase cells dropped as intermediate and each state’s response referred to its own phase-matched control. The second is k -means with $k = 2$ fitted on each cell line’s DMSO_TF control cells in the 2,304 highly variable genes in log1p space, with ten initializations and a fixed seed; treated cells are assigned to the nearest control centroid, the larger control cluster is taken as the first state, and a cell line enters only when it holds at least 100 control cells. Across 3,630 evaluable drug-cell-line conditions, the median cosine between the two state-specific response directions was 0.739 (interquartile range 0.600–0.836). Fewer than 0.11% of conditions were as divergent as the constructed mixtures used in the controlled experiments.

This estimate is not driven materially by the minimum-cell filter. Conditions were excluded when one of the required state-specific treated or control arms contained fewer than 50 cells, which can preferentially remove strong cell-cycle-arresting perturbations. Across drugs, coverage was only weakly associated with induced-response cosine (Spearman $\rho = 0.077$), and restricting the analysis to well-covered compounds changed the median from 0.739 to 0.740.

Recoverability remained measurable at scale. Across 960 sampled drug pairs and 48 cell lines, the supervised ceiling was 0.853 and the best unsupervised procedure reached 0.611, giving a median paired gap of 0.157. The gap was positive in all 48 evaluable cell lines. Individual unsupervised methods gave similar results: k -means 0.592, Gaussian mixtures 0.582 and Leiden¹ 0.563.

State-specific ordering and decision-relevance headroom. We finally compared relative drug-response ordering between non-overlapping cellular states. Under the cell-cycle partition, the median disjoint Spearman correlation between the two drug-drug similarity structures was 0.841 across 44 evaluable contexts. Under the independently control-derived two-state partition, the corresponding median was 0.778 across 45 contexts.

These high correlations indicate that relative drug ordering is often substantially preserved between cellular states even when the states respond differently. The pattern is not universal: under the control-derived partition, 2 of 45 contexts had a correlation below 0.5, with a minimum of 0.236. Such contexts leave greater headroom for state-specific response information to alter candidate ordering. Headroom of that kind is what a population-level method would need in order to change the selected candidate, and it is measured here on the ordering alone.

The two state definitions agree only moderately in their ranking of contexts (Spearman $\rho = 0.466$), reinforcing that apparent decision-relevance headroom depends on how cellular states are defined. Both partitions were specified independently of downstream retrieval performance.

Tahoe-100M therefore supports the same separation of concepts observed elsewhere in the study: state-specific responses can exist, and response identity can remain recoverable, while much of the relative candidate ordering is still preserved across cellular states. These are intrinsic-state diagnostics: they constrain the conditions under which population-level retrieval could add value. The advantage itself

is measured on the same plate by the source-to-target intervention benchmark of the main text (Figs. 5 and 6).

Supplementary Note 4. Perturbation-predictor diagnostics

The predictor analyses ask whether a predicted response population contains state-specific perturbation effects that a downstream population-aware retriever could exploit.

Additive response models. For a predictor that applies the same perturbation effect δ_d to every cell in a context,

$$\hat{x}_{i,d} = x_i + \delta_d,$$

the response of any two cellular states is identical after subtraction of their corresponding controls,

$$r_{d,0} = r_{d,1} = \delta_d.$$

Consequently,

$$\cos(r_{d,0}, r_{d,1}) = 1.$$

This result is independent of the heterogeneity of the starting control population. An additive predictor can therefore generate a visually heterogeneous predicted population while containing no state-specific variation in the perturbation effect itself.

The average-effect and linear-latent predictors used in the main analysis satisfy this additive limit by construction when the perturbation response is referenced to the model-consistent baseline. They establish this limiting case.

The predict-then-rank construction. The predictor-generated retrieval factorizes the pipeline into a forward predictor and a retrieval rule, with the retrieval layer varied while the predictor is held fixed. It is run on SciPlex3 and is separate from the Tahoe-100M prediction arm of the main text. Queries are cross-line: for each of the three SciPlex3 cell-line pairs (K562–A549, A549–MCF7, K562–MCF7) and each of ten drugs selected by a cross-context divergence probe, the query is that drug’s response blended across the two lines at mixing fraction $\alpha \in \{0.5, 0.7, 0.9\}$ over 200 cells, with eight seeds, giving 720 queries. The candidate library is the query’s own drug together with up to 30 further drugs shared by both lines; the query’s drug is the single relevant candidate. Each predictor is asked to imagine every candidate’s response in the query’s real blended control cells, mixed at the same α , so no predictor sees the within-query line structure. The imagined populations are then ranked by four retrieval rules: mean cosine, R^2 of the predicted against the observed mean delta, energy distance and coverage. Performance is nDCG@10 with binary relevance, and the reported quantity is the distributional minus the mean-cosine value for the same predictor. The predictors are fitted on the full SciPlex3 matrix, including the lines that compose each query, so these values are an in-sample ceiling on candidate generation.

Two reference baselines. A predicted population’s induced response can be referred to two different controls, and the two answer different questions. Referring it to the model’s own zero-effect baseline, which is its reconstruction of the control population whenever the model decodes and the control cells themselves when it does not, cancels the reconstruction bias and leaves the divergence the model itself generates. The additivity result above pins this quantity to exactly 1 for any predictor of the form $\hat{x}_{i,d} = g(x_i) + \delta_d$ with a fixed g , covering both the gene-space case $g = \text{id}$ and the latent case in which g is the model’s own reconstruction, so the value is analytic and not estimated. Referring it to the real control cells is the retrieval-scorer convention, and the resulting cosine contains model reconstruction bias, which differs between cellular states. Across the same 30 cell-line and drug combinations, the average-effect, linear-latent, scGen and optimal-transport predictors give 1.000, 1.000, 0.972 and 0.900 against the model baseline, and 1.000, 0.190, 0.404 and 0.380 against the real control; real treated cells give 0.205, and have no model baseline.

The two columns are informative together. Lower cosine means more state-specific divergence. The average-effect predictor adds its effect directly in gene space and has no reconstruction step, so its two

286 references coincide at 1.000. The other three pass their output through a decoder whose reconstruction
 287 error differs between cellular states, and that error enters the real-control cosine as a perturbation compo-
 288 nent. The linear-latent model generates exactly zero state-specific divergence, yet its real-control cosine
 289 is 0.190, compared with 0.205 for the real cells it replaces. A distributional score on such a predicted
 290 population therefore includes decoder effects.

291 **scGen.** A deployed scGen model was evaluated using the same within-context cellular-state partition
 292 and state-specific induced-response definition as the other predictors. Across the same 30 cell-line and
 293 drug combinations, its mean induced-response cosine against its own reconstruction baseline was 0.972,
 294 close to the additive limit of 1.000, and 0.404 against the real control. Thus, in this experiment scGen
 295 preserved substantially less state dependence in the perturbation effect than the 0.205 observed in real
 296 treated cells, under either reference.

297 The model is the published `scgen`.SCGEN implementation on `scvi-tools`³, trained on the SciPlex3
 298 tensor with the perturbation label as batch key and the cell line as label key, 30 latent dimensions, batch
 299 size 512, at most 40 epochs with early stopping after 15 epochs without improvement, and a fixed seed.
 300 Predictions use scGen’s own latent arithmetic: one latent delta per perturbation, taken as the mean latent
 301 of treated cells minus the mean latent of control cells, added to the encoded control mean and decoded.

302 **Entropic optimal-transport mapping.** We additionally evaluated an entropic optimal-transport map as
 303 a predictor with cell-dependent displacements. It is a different map from the Gaussian (Bures) predictor
 304 of the Tahoe-100M arm (Methods). Under the within-context differential-response analysis, the optimal-
 305 transport predictions reached an induced-response cosine of approximately 0.900 ± 0.107 against their
 306 own reconstruction baseline, moving away from the additive limit but remaining much more aligned
 307 than the observed responses.

308 A latent-space implementation passed the corresponding perturbation-learning diagnostic (mean-
 309 delta agreement 0.352), whereas the same construction applied directly in gene space showed substan-
 310 tially weaker perturbation learning (0.057). The latent-space implementation was therefore used for the
 311 differential-response comparison.

312 The map is an entropic (Sinkhorn) coupling⁴ between control and treated cells, solved with POT⁵,
 313 with uniform marginals, regularization 0.05, at most 500 iterations and a cost matrix rescaled by its
 314 maximum, solved in a 30-component PCA latent fitted unsupervised on at most 20,000 cells. Each fitting
 315 context is centred on its own control mean, and the query context is excluded from the fit. A barycentric
 316 projection gives a per-source-cell displacement, extended to the query’s real control cells by nearest-
 317 source matching in the same centred space and decoded through the PCA inverse transform.

318 In the cross-context predict-then-rank setting, the same entropic map gave Hit@1 values of 0.053
 319 under mean cosine and 0.007 and 0.008 under the two distributional rules, compared with 0.879 to 0.899
 320 for the additive average-effect candidate source. The predictions therefore did not preserve retrieval
 321 performance in that transfer setting.

322 Taken together, these analyses identify a specific next experiment. A decisive test requires a pre-
 323 dictor that both generalizes accurately to the target cellular context and reproduces the state-specific
 324 perturbation responses present in real cells. No predictor evaluated here satisfied both requirements
 325 simultaneously.

Supplementary Table 1. Retrieval scores, task constructions and evaluation criteria.

(a) Retrieval scores.

Score	Input representation	Ranking rule	Role
Mean cosine	Control-referenced mean perturbation signatures	Descending cosine similarity between query and candidate mean-response vectors	Primary mean-signature baseline
CMap-style cosine	Same control-referenced mean differential-expression vectors as mean cosine	Same cosine operation as the implemented mean-cosine baseline	Signature-retrieval reference; numerically identical to mean cosine in this implementation

Supplementary Table 1(a) continued.

Score	Input representation	Ranking rule	Role
WTCS	Mean-collapsed differential-expression response	Rank-based weighted connectivity / enrichment score	Canonical rank-based signature baseline; related to but not algebraically identical to mean cosine
Energy distance	Raw normalized-expression single-cell response populations	Candidates ranked by increasing U-statistic energy distance to the query population	Primary population-level retrieval score
RBF-MMD	Raw normalized-expression response populations	Candidates ranked by increasing U-statistic maximum mean discrepancy under an RBF kernel	Alternative distribution-sensitive score
Sliced Wasserstein	Empirical response populations	Candidates ranked by increasing sliced Wasserstein distance using 128 random projections	Alternative distribution-sensitive score
Subpopulation coverage	Response populations decomposed into response components	Component-level distances aggregated with temperature β ; increasing β gives progressively greater weight to the least well-matched response component	Population-sensitive score used for the partial-observation response-matching analyses

326 The four score families become five population-level configurations because subpopulation-coverage contributes two:
327 coverage-mean is the $\beta \rightarrow 0$ limit of the aggregate and coverage-worst the $\beta \rightarrow \infty$ limit, both with energy distance as the
328 per-component base metric and the same U-statistic default for energy/MMD components. The CMap-style cosine is numeri-
329 cally identical to mean cosine on control-referenced mean vectors. Its sign-flipped variant is scored in the partial-observation
330 experiments as an anti-correlation floor. Sliced Wasserstein has no U/V estimator choice; the PCA-latent V arm is legacy-only.

Supplementary Table 1 continued. (b) Retrieval task constructions.

Task	Query and candidate construction	Candidate evaluation	Statistical unit
Observed-population retrieval over heterogeneous queries	A heterogeneous query population combines two response modes at weight α ; the library holds a held-out re-mixture at the same weight, the two single-mode populations, and 40 distractors, giving a 43-candidate library	Hit@1; only the both-mode candidate is relevant, so a single-mode population counts as a miss even when it carries one of the query's two response modes	Retrieval query; headline values macro-averaged over seven task-setting combinations
Controlled heterogeneous queries	Two response modes are combined at mixing weight α ; controlled SciPlex3 analyses use HDAC- and JAK-associated response sources	Source identification and response-matching criteria	Query / task setting
Partial-observation retrieval	Only the response information available under the controlled partial-observation protocol is exposed to the retriever; population and mean methods rank the identical candidate library once	The resulting fixed ranking is evaluated under response matching, MoA recovery and minority-state coverage	765 total queries; formal inference clusters 720 individual-drug queries by 143 query drugs; 45 pooled partial-library queries are descriptive only
Mechanism-evaluable subset	Same partial-observation rankings; restricted to queries for which the query mechanism remains represented among candidates	MoA-nDCG	600 leave-drug-out queries from 130 query-drug clusters; 120 leave-MoA-out and 45 partial-library queries are excluded because MoA recovery is undefined

Supplementary Table 1(b) continued.

Task	Query and candidate construction	Candidate evaluation	Statistical unit
Real-data minority-state projection	Top-ranked population and mean candidates evaluated on 239 task instances spanning constructed cross-line, within-line, Frangieh, CD34+ and predicted-population settings	Difference in minority-state coverage	Task-level difference across 239 task instances
GDSC2-linked retrieval	Held-out SciPlex3 compound at 10 μ M used as query; remaining matched compounds form a 34–35 candidate library	Spearman agreement between RNA-derived ranking and external GDSC2 functional-similarity ordering	Per-query correlation summarized within SciPlex3 cell line

Supplementary Table 1 continued. (c) Evaluation criteria applied to fixed rankings.

Criterion	Definition	Interpretation
Hit@1	Indicator that the both-mode candidate, the held-out re-mixture at the query’s own weight, is ranked first	Recovery of the query’s response composition, not of its perturbation identity alone
Response-matching regret	Energy-based welfare regret against the latent per-cell objective used to construct the controlled partial-observation task; reported as regret reduction relative to mean retrieval	Response fidelity; intentionally close to the population information used by the retriever
MoA-nDCG	Normalized discounted cumulative gain rewarding candidates that share the query perturbation’s annotated mechanism	Task-proximal biological criterion; undefined when the relevant mechanism is absent from the candidate library
Minority-state coverage	Measures whether the selected candidate better preserves the response of the less-represented cellular state	Task-proximal measure of coverage of heterogeneous response structure
GDSC2 functional similarity	Spearman correlation between cell-line-centred cross-cell-line AUC profiles of the query and candidate drugs	Externally measured functional similarity; not absolute potency
Protein mean evaluator	Cosine similarity between control-referenced mean protein-response vectors	External evaluator preserving only mean protein response
Protein population evaluator	Negative energy distance between control-referenced per-cell protein-response populations	External evaluator retaining cell-to-cell protein-response structure
Response-magnitude matching	$-\ \vec{d}_q\ - \ \vec{d}_c\ $	Query-dependent scalar control that uses no within-population information

Query states s are obtained by k -means with $k = 2$ on the query cells, the minority state being the smaller cluster. A candidate d with response population P_d receives the welfare $U(d) = \text{agg}_s(-\mathcal{E}(P_d, Q_s))$, where $\mathcal{E}$ is the energy distance and the aggregator is the minimum over states in the worst-case form and the mean in the mean form. The response-matching regret of a selected candidate $\hat{d}$ is $\text{Regret}(\hat{d}) = \max_{d \in \mathcal{L}} U(d) - U(\hat{d}) \geq 0$, taken over the observed library $\mathcal{L}$, so the reference is the best available candidate in the library. Minority-state coverage is $\frac{1}{2}(1 + \cos(\bar{P}_d, \bar{Q}_{\min}))$, the cosine between the candidate’s mean response and the mean response of the query’s minority state rescaled to $[0, 1]$. The same definitions are given in the main Methods.

Supplementary Table 2. Data resources, preprocessing and analytical roles.

Resource	Scale	Perturbation/context structure	Representation and controls	Role in this study
SciPlex3	276,325 cells; 2,000 genes	188 compounds shared across A549, K562 and MCF7	Log-transformed 2,000-gene representation; mean signatures are referenced to matched controls, whereas the manuscript population scorers use normalized treated-cell populations	Primary observed-population drug-retrieval setting; controlled within- and cross-context constructions; source of the GDSC2-linked transcriptomic rankings

Supplementary Table 2 continued.

Resource	Scale	Perturbation/context structure	Representation and controls	Role in this study
Frangieh Perturb-CITE-seq	218,023 RNA-profiled cells for retrieval and 218,331 barcode-matched cells for the RNA-protein evaluation; 2,000 genes; 20 retained surface proteins	239 CRISPR knock-outs across Control, IFN γ and Co-culture conditions	RNA and protein modalities joined by cell barcode; mean signatures are referenced to condition-matched controls, whereas the manuscript population scorers use normalized treated-cell populations	Independent genetic-perturbation retrieval setting and matched RNA-protein evaluator-shape experiment
CD34+ HSPC	33,984 cells; 2,000 genes	36 compounds	Log-transformed expression; DMSO-referenced perturbation responses	Negative-control retrieval setting for testing whether population-level information consistently improves ranking
ZhaoSims2021 glioblastoma	10 patients; 6 compounds	6 Acute-slice culture and surgical biopsy with co-existing malignant, myeloid and oligodendrocyte compartments	Counts normalized to 10 ⁴ per cell, log1p-transformed and reduced to 2,000 highly variable genes; state-specific responses referenced within patient and compartment	Natural-tissue analysis of differential response, recoverability and state-specific response ordering
Tahoe-100M plate 3	4,158,278 cells; 2,304 genes	50 cell lines $\times$ 93 compounds at one concentration, with DMSO_TF controls	Released 2,304-gene log1p representation used without refitting; state-specific responses referenced to matched state controls	Two roles: large-scale measurement of differential response, recoverability and state-ordering agreement under intrinsic within-cell-line heterogeneity, and the substrate for the source-to-target intervention retrieval benchmark (Figs. 5 and 6)
GDSC2	286 compounds across 969 cell lines in the retained release	35 compounds matched to the SciPlex3 panel; A549, K562 and MCF7 excluded from external profiles	Fitted dose-response AUC; primary functional similarity uses cell-line-centred cross-line response profiles	External functional evaluation of fixed SciPlex3 RNA-derived rankings
HIR-Bench	13,440 synthetic instances	672 parameter configurations $\times$ 20 seeds in a two-subpopulation latent-welfare model	Synthetic observed, predicted-mean and predicted-structure information conditions	Controlled demonstration that population structure can alter candidate preference

337 The resources above address distinct questions. Dataset-specific cell-number thresholds, subsampling rules and statistical units
338 are given in the corresponding Methods sections.

Supplementary Table 3. Sensitivity of the GDSC2 functional-similarity evaluation to response-profile construction.

The primary external functional criterion compares drug–drug response profiles after removing the shared cell-line sensitivity effect. For cell line l and drug d ,

$$\widetilde{\text{AUC}}_{d,l} = \text{AUC}_{d,l} - \frac{1}{|\mathcal{D}_l|} \sum_{d' \in \mathcal{D}_l} \text{AUC}_{d',l},$$

where $\mathcal{D}_l$ contains all GDSC2 compounds measured in that cell line. Drug–drug functional similarity is then the Spearman correlation between centred cross-cell-line response profiles. The table compares this primary construction with the corresponding uncentred AUC-profile similarity.

RNA-derived ranking rule	Cell-line-centred functional similarity	Uncentred functional similarity
Energy distance	+0.265	+0.0902
Mean cosine, control referenced	+0.083	+0.138
Mean cosine, treated mean only	+0.241	+0.092
Response-magnitude matching	+0.232	+0.217
Candidate magnitude only	−0.330	+0.135
Potency matching diagnostic	+0.399	+0.290
Energy, magnitude-adjusted partial rank correlation	+0.105	n/a

339 Values are median per-query Spearman correlations over 103 leave-one-drug-out queries across the three SciPlex3 cell lines. The
340 primary centred functional criterion is constructed from GDSC2 AUC profiles after subtracting each cell line’s mean response
341 across all 286 compounds in the retained release. Drug pairs sharing fewer than 100 cell lines are excluded; 591 usable drug
342 pairs remain, with a median of 909 shared cell lines per pair.

343 The change between centred and uncentred constructions shows that external evaluation is itself sensitive to how the functional
344 measurement is summarized. Under the centred criterion, energy retrieval exceeds the control-referenced mean-signature
345 baseline, but much of this association is also reproduced by response-magnitude matching. Under the uncentred criterion, the
346 ordering between energy and the mean-signature baseline reverses. Neither construction is interpreted as a direct measure of
347 therapeutic utility.

Supplementary Table 4. Perturbation-predictor implementations and their role in the differential-response analysis.

Predictor	Model form	Predicted-population construction	Use in this study	Interpretation / limitation
Average-effect	Per-drug mean perturbation delta	Predicted delta applied to the real control cells of the query context	Predict-then-rank candidate source and within-context differential-response reference	Additive by construction; state-specific induced-response cosine equals 1
Nearest-neighbour	Transfers the perturbation effect from the training context with the most similar untreated control state	Transferred effect applied to real query-context control cells	Predict-then-rank candidate source	Tests simple context transfer
Linear-latent	Additive latent arithmetic in an encoded representation followed by linear decoding	Observed control cells encoded, shifted in latent space and decoded individually	Predict-then-rank candidate source and additive latent reference	Can preserve baseline cellular heterogeneity while remaining additive in perturbation effect
scGen	Variational-autoencoder perturbation model with latent arithmetic	Evaluated on the same within-context state partition and state-specific response definition used for other predictors	Additional differential-response diagnostic reported in the main text	Mean induced-response cosine 0.972 over 30 cell-line and drug combinations

Supplementary Table 4 continued.

Predictor	Model form	Predicted-population construction	Use in this study	Interpretation / limitation
Entropic optimal-transport map	Non-parametric Sinkhorn mapping that permits cell-dependent displacements	Control cells mapped individually to predicted treated states	Additional non-additive differential-response diagnostic; also tested as a candidate-response source	Within-context induced-response cosine approximately 0.900 ± 0.107 against the map's own reconstruction baseline and 0.380 against the real control; cross-context candidate predictions were too inaccurate for a clean retrieval test

Supplementary Table 4 continued. Implementation of each predictor.

Predictor	Implementation
Average-effect	Per drug d , $\hat{\delta}_d = \text{mean}_c (\text{mean}(\text{treated}_{c,d}) - \text{mean}(\text{control}_c))$ over the SciPlex3 cell lines c , a line contributing only where it has at least two treated cells for d ; one signature per drug, independent of the query context. The predicted population is that single signature added to each of the query context's real control cells, so every cell is shifted identically. No reconstruction step, so the model's own zero-effect baseline is the control cells themselves. A drug absent from the fit is scored at -10^6 under every rule rather than imputed
Nearest-neighbour	$c^* = \arg \min_c (1 - \cos(\bar{x}_{\text{control}}^{\text{query}}, \bar{x}_{\text{control}}^c))$ over the training lines, then $\hat{\delta}_d = \text{mean}(\text{treated}_{c^*,d}) - \text{mean}(\text{control}_{c^*})$. Distances are compared strictly, so an exact tie is resolved by context iteration order; where d was not measured in c^* the signature falls back to the mean of d 's signatures over the remaining admissible lines. Population synthesis as for average-effect. The two query-composing cell lines remained eligible donor contexts, so this experiment is interpreted as an in-sample ceiling rather than as a transfer evaluation
Linear-latent	PCA with 30 components fitted, unsupervised and once, on up to 20,000 cells drawn from the full SciPlex3 matrix at seed 0. Each line's control latent mean is stored; each drug's latent delta is the mean over lines of (mean latent of treated – that line's control latent mean), again requiring at least two treated cells. Prediction encodes each real control cell of the query context, adds the drug's latent delta to that cell's own latent code, and decodes each cell by the PCA inverse transform. The decoder is linear, so the composite map is $\text{dec}(\text{enc}(x) + dz) = Px + Pdz$ and the predictor is additive in gene space and deterministic. Because it reconstructs, its own zero-effect baseline is the PCA reconstruction of the control cells rather than those cells
scGen	Published <code>scgen</code> .SCGEN on <code>scvi-tools</code> ; trained on the SciPlex3 tensor, batch key = perturbation, label key = cell line; 30 latent dimensions; batch size 512; at most 40 epochs with early stopping after 15 epochs without improvement; fixed seed; per-perturbation latent delta added to the encoded control mean and decoded
Optimal transport	Entropic (Sinkhorn) coupling in a 30-component PCA latent; cost rescaled by its maximum; uniform marginals; regularization 0.05; at most 500 iterations; fitted per non-query context centred on that context's own control mean, with the query context excluded; barycentric projection extended to the query's control cells by nearest-source matching and decoded by the PCA inverse transform

348 Predictors are used as candidate-response sources or information diagnostics rather than as a benchmark of perturbation-
349 prediction architectures. The reported predict-then-rank experiment uses predictors fitted on the full SciPlex3 matrix and is
350 therefore interpreted as an in-sample ceiling, not as a cross-context generalization benchmark.

351 Constants shared by every row of the predict-then-rank run: 720 cross-line queries, formed from the three cell-line pairs (K562–
352 A549, A549–MCF7, K562–MCF7) at mixing weights $\alpha \in \{0.5, 0.7, 0.9\}$ over 10 divergent drugs and 8 seeds; 200 query cells
353 per query; a 31-drug candidate library holding the true drug and 30 other shared drugs; 200 predicted cells synthesized per
354 candidate; query construction seeded at $1000 + s$ and distractor selection at $7000 + s$ for $s = 0, \dots, 7$. The within-context
355 differential-response measurements in Supplementary Note 4 are a separate run over 30 cell-line and drug combinations with
356 400 control cells per context split into two states, a state requiring 60 cells and a treated arm 40, at seed 0. Nearest-neighbour
357 appears only in the predict-then-rank run and scGen only in the differential-response run; the entropic optimal-transport map
358 appears in both.

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